

Problem

The 3N+1 sequence is generated using the following rules: If n is even, the next number is n / 2.If n is odd, the next number is 3 × n + 1.The sequence continues until it reaches 1. The cycle length of a number is the total number of terms in the sequence, including the starting number and 1.

Given two integers i and j, determine the maximum cycle length among all integers between i and j (inclusive).

If i > j, swap them before processing.

Input Format

A single line containing two space-separated integers:

i j

Constraints

1 ≤ i, j ≤ 1,000,000

Output Format

Print three integers:

i j maximum_cycle_length

using the original values of i and j.

Sample Input 0

1 10

Sample Output 0

1 10 20

Submissions

Leaderboard

Discussions

Change Theme

Language Java 15

```
1 import java.util.*;
2
3 public class Solution {
4     public static int cycleLength(long n) {
5         int count = 1;
6         while (n != 1) {
7             if (n % 2 == 0) {
8                 n = n / 2;
9             } else {
10                 n = 3 * n + 1;
11             }
12             count++;
13         }
14         return count;
15     }
16     public static void main(String[] args) {
17         Scanner sc = new Scanner(System.in);
18
19         int i = sc.nextInt();
20         int j = sc.nextInt();
21         int originalI = i;
22         int originalJ = j;
23         if (i > j) {
24             int temp = i;
25             i = j;
```

Line: 5 Col: 23

Upload Code as File

Run Code

Submit Code

☐ Test against custom input

Congratulations

You solved this challenge. Would you like to challenge your friends?

Test case 0

Test case 1

Test case 2

Compiler Message

Success